THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS CHAPTER TEST SET B

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number 1 is: 1

(a) neither prime nor composite **(b)** composite
(c) even **(d)** prime

2. Two numbers whose only common factor is 1 are: 1

(a) twin primes **(b)** co-prime **(c)** composite **(d)** equal

3. A number is divisible by 2 if its last digit is: 1

(a) 1, 3, 5, 7, 9 **(b)** 0, 2, 4, 6 or 8 **(c)** any digit **(d)** 0 or 5 only

4. A number is divisible by 10 if its last digit is: 1

(a) 5 **(b)** any **(c)** 1 **(d)** 0

5. The number of primes between 1 and 20 is: 1

(a) 7 **(b)** 8 **(c)** 10 **(d)** 9

6. The HCF is the: 1

(a) highest common multiple **(b)** highest common fraction
(c) half of the common factor **(d)** highest common factor

7. The smallest 2-digit prime is: 1

(a) 17 **(b)** 10 **(c)** 13 **(d)** 11

8. The HCF of 12 and 18 is: 1

(a) 7 **(b)** 12 **(c)** 6 **(d)** 5

9. The HCF of 16 and 24 is: 1

(a) 16 **(b)** 7 **(c)** 9 **(d)** 8

10. The LCM of 4 and 6 is: 1

(a) 12 **(b)** 6 **(c)** 16 **(d)** 48

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Find the LCM of 4 and 10. 2

12. What are co-prime numbers? Give an example. 2

13. Is 91 prime? Give reason. 2

14. Write the prime factorisation of 36. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. A rope of 48 m and another of 36 m are cut into equal pieces of the longest possible length. Find the length. 3

16. Three bells ring every 6, 8 and 12 minutes. After how many minutes will they ring together? 3

17. Find the factors of 36 and state how many there are. 3

OR

Express 90 as a product of primes. How many prime factors does it have? 3

18. Are 35 and 64 co-prime? Find their HCF. 3

OR

State and test the divisibility rule of 9 on 7,128. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) List the prime numbers from 1 to 50. 5
(ii) How many are there?
(iii) Find the twin primes among them.

OR

(i) Explain co-prime numbers. 5
(ii) Which pairs are co-prime: (21, 22), (18, 27), (35, 36), (14, 49)?
(iii) Give a real-life use.

20. **Case study:** Two traffic lights turn green every 40 s and 60 s starting together. 5
(i) After how many seconds do they turn green together again?
(ii) How many times in 10 minutes?
(iii) Which concept is used?
(iv) If the periods were 7 s and 9 s?

OR

(i) Explain the Sieve of Eratosthenes. 5
(ii) List primes up to 30 obtained.

END OF QUESTION PAPER Total: 40 Marks

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